Accelerator ML Living Review

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Papers

Machine learning as a service system for particle accelerator and its application in CSNS
Mei, Hao, Zhang, Yuliang, Peng, Na, Cheng, Sinong, He, Yongcheng, Xue, Kangjia, Wang, Lin, Li, Mingtao, Wu, Xuan, Zhu, Peng (2025)
Radiation Detection Technology and Methods

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry
Asif Iqbal, John Verboncoeur, Peng Zhang (2025)
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Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls
Madysa, Penelope, Appel, Sabrina, Kain, Verena, Schenk, Michael (2025)
SoftwareX

Acceleration of Multi-Scale LTS Magnet Simulations with Neural Network Surrogate Models
Louis Denis, Julien Dular, Vincent Nuttens, Mariusz Wozniak, Benoît Vanderheyden, Christophe Geuzaine (2025)
arXiv

Application Of Large Language Models For The Extraction Of Information From Particle Accelerator Technical Documentation
Qing Dai, Rasmus Ischebeck, Maruisz Sapinski, Adam Grycner (2025)
arXiv

Machine Learning approach to classifying quench antenna signals
Plebani, Alberto, Barzi, Emanuela, Teyber, Reed (2025)
Thesis

Machine Learning for superconducting magnets application
Stabilini, Elisa (2025)
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Batch spacing optimization by reinforcement learning
Remta, Matthias, Velotti, Francesco, Rezagholi, Sharwin (2025)
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Machine learning application for particle accelerator optimization-a review
Isti Dian Rachmawati, Nazrul Effendy, Taufik Taufik (2025)
IAES International Journal of Artificial Intelligence

Machine Learning for Particle Accelerators
Elena Fol, Auralee Edelen (2025)
Unknown Venue

Machine learning based parametrization of the resolution function for the first experimental area of the n_TOF facility at CERN
Petar Žugec, Marta Sabaté-Gilarte, Michael Bacak, Vasilis Vlachoudis, Adria Casanovas, Francisco García-Infantes (2025)
Nuclear Science and Techniques

Integration of Machine Learning-Based Plasma Acceleration Simulations into Geant4: A Case Study with the PALLAS Experiment
A. Sytov, K. Cassou, V. Kubytskyi, M. Lenivenko, A. Huber (2025)
arXiv

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach
Anwar Ibrahim, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin (2025)
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Physics-Informed Super-Resolution Diffusion for 6D Phase Space Diagnostics
Alexander Scheinker (2025)
arXiv

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations
Hwang, Robin (2024)
arXiv

Virtual Pulse Reconstruction Diagnostic for Single-Shot Measurement of Free Electron Laser Radiation Power
Till Korten, Vladimir Rybnikov, Peter Steinbach, Najmeh Mirian (2024)
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Harnessing Machine Learning for Single-Shot Measurement of Free Electron Laser Pulse Power
Korten, Till, Rybnikov, Vladimir, Vogt, Mathias, Roensch-Schulenburg, Juliane, Steinbach, Peter, Mirian, Najmeh (2024)
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Data-driven gradient optimization for field emission management in a superconducting radio-frequency linac
Goldenberg, Steven, Ahammed, Kawser, Carpenter, Adam, Li, Jiang, Suleiman, Riad, Tennant, Chris (2024)
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Data-Driven Discovery of Beam Centroid Dynamics
Pocher, Liam A., Haber, Irving, Antonsen, Thomas M., O'Shea, Patrick G. (2024)
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Design and development of advanced Al-Ti-V alloys for beampipe applications in particle accelerators
Kamaljeet Singh, Kangkan Goswami, Raghunath Sahoo, Sumanta Samal (2025)
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Machine Learning for Reducing Noise in RF Control Signals at Industrial Accelerators
M. Henderson, J. P. Edelen, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2024)
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Beamline Steering Using Deep Learning Models
Allen, Dexter, Kante, Isaac, Bohler, Dorian (2024)
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Beam-based Identification of Magnetic Field Errors in a Synchrotron using Deep Lie Map Networks
Conrad Caliari, Adrian Oeftiger, Oliver Boine-Frankenheim (2024)
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Linac_Gen: integrating machine learning and particle-in-cell methods for enhanced beam dynamics at Fermilab
Abhishek Pathak (2024)
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Automated Anomaly Detection on European XFEL Klystrons
Antonin Sulc, Annika Eichler, Tim Wilksen (2024)
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Accelerator beam phase space tomography using machine learning to account for variations in beamline components
Andrzej Wolski, Diego Botelho, David Dunning, Amelia E. Pollard (2024)
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Large Language Models for Human-Machine Collaborative Particle Accelerator Tuning through Natural Language
Jan Kaiser, Annika Eichler, Anne Lauscher (2024)
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Accelerating Cavity Fault Prediction Using Deep Learning at Jefferson Laboratory
Monibor Rahman, Adam Carpenter, Khan Iftekharuddin, Chris Tennant (2024)
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Anomaly Detection of Particle Orbit in Accelerator using LSTM Deep Learning Technology
Zhiyuan Chen, Wei Lu, Radhika Bhong, Yimin Hu, Brian Freeman, Adam Carpenter (2024)
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Machine-learning approach for operating electron beam at KEK e^-/e^+ injector Linac
Gaku Mitsuka, Shinnosuke Kato, Naoko Iida, Takuya Natsui, Masanori Satoh (2024)
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Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations
Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia (2024)
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Beyond PID Controllers: PPO with Neuralized PID Policy for Proton Beam Intensity Control in Mu2e
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Robust Errant Beam Prognostics with Conditional Modeling for Particle Accelerators
Kishansingh Rajput, Malachi Schram, Willem Blokland, Yasir Alanazi, Pradeep Ramuhalli, Alexander Zhukov, Charles Peters, Ricardo Vilalta (2024)
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Machine Learning For Beamline Steering
Isaac Kante (2023)
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Variational Autoencoders for Noise Reduction in Industrial LLRF Systems
J. P. Edelen, M. J. Henderson, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2023)
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Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source
Ryan Humble, William Colocho, Finn O'Shea, Daniel Ratner, Eric Darve (2023)
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Time-drift Aware RF Optimization with Machine Learning Techniques
R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2023)
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Distance Preserving Machine Learning for Uncertainty Aware Accelerator Capacitance Predictions
Steven Goldenberg, Malachi Schram, Kishansingh Rajput, Thomas Britton, Chris Pappas, Dan Lu, Jared Walden, Majdi I. Radaideh, Sarah Cousineau, Sudarshan Harave (2023)
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Learning to Do or Learning While Doing: Reinforcement Learning and Bayesian Optimisation for Online Continuous Tuning
Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia, Oliver Stein, Erik Bründermann, Willi Kuropka, Hannes Dinter, Frank Mayet, Thomas Vinatier, Florian Burkart, Holger Schlarb (2023)
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Forecasting Particle Accelerator Interruptions Using Logistic LASSO Regression
Sichen Li, Jochem Snuverink, Fernando Perez-Cruz, Andreas Adelmann (2023)
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Learning Electron Bunch Distribution along a FEL Beamline by Normalising Flows
Anna Willmann, Jurjen Couperus Cabada■, Yen-Yu Chang, Richard Pausch, Amin Ghaith, Alexander Debus, Arie Irman, Michael Bussmann, Ulrich Schramm, Nico Hoffmann (2023)
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Physics-constrained 3D Convolutional Neural Networks for Electrodynamics
Alexander Scheinker, Reeju Pokharel (2023)
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Identification of Magnetic Field Errors in Synchrotrons based on Deep Lie Map Networks
Conrad Caliari, Adrian Oeftiger, Oliver Boine-Frankenheim (2023)
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Data-driven Science and Machine Learning Methods in Laser-Plasma Physics
Andreas Döpp, Christoph Eberle, Sunny Howard, Faran Irshad, Jinpu Lin, Matthew Streeter (2023)
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Applications of Differentiable Physics Simulations in Particle Accelerator Modeling
Ryan Roussel, Auralee Edelen (2022)
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Prior-mean-assisted Bayesian optimization application on FRIB Front-End tunning
Kilean Hwang, Tomofumi Maruta, Alexander Plastun, Kei Fukushima, Tong Zhang, Qiang Zhao, Peter Ostroumov, Yue Hao (2022)
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Fault Prognosis in Particle Accelerator Power Electronics Using Ensemble Learning
Majdi I. Radaideh, Chris Pappas, Mark Wezensky, Pradeep Ramuhalli, Sarah Cousineau (2022)
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Machine learning-based analysis of experimental electron beams and gamma energy distributions
M. Yadav, M. Oruganti, S. Zhang, B. Naranjo, G. Andonian, Y. Zhuang, Ö. Apsimon, C. P. Welsch, J. B. Rosenzweig (2023)
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Review of Time Series Forecasting Methods and Their Applications to Particle Accelerators
Sichen Li, Andreas Adelmann (2022)
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Automatic setup of 18 MeV electron beamline using machine learning
Francesco Maria Velotti, Brennan Goddard, Verena Kain, Rebecca Ramjiawan, Giovanni Zevi Della

Porta, Simon Hirlaender (2022)
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Diagnostics for Linac Optimization With Machine Learning
R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2022)
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Transverse phase space tomography in the CLARA accelerator test facility using image compression and machine learning
Andrzej Wolski, Mark A. Johnson, Matthew King, Boris L. Militsyn, Peter H. Williams (2022)
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Using Kernel-Based Statistical Distance to Study the Dynamics of Charged Particle Beams in Particle-Based Simulation Codes
Chad E. Mitchell, Robert D. Ryne, Kilean Hwang (2022)
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Neural Network Solver for Coherent Synchrotron Radiation Wakefield Calculations in Accelerator-based Charged Particle Beams
Auralee Edelen, Christopher Mayes (2022)
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Adaptive Machine Learning for Time-Varying Systems: Towards 6D Phase Space Diagnostics of Short Intense Charged Particle Beams
Alexander Scheinker, Spencer Gessner (2022)
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Differentiable Preisach Modeling for Characterization and Optimization of Accelerator Systems with Hysteresis
R. Roussel, A. Edelen, D. Ratner, K. Dubey, J. P. Gonzalez-Aguilera, Y. K. Kim, N. Kuklev (2022)
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Mixed Diagnostics for Longitudinal Properties of Electron Bunches in a Free-Electron Laser
J. Zhu, N. M. Lockmann, M. K. Czwalinna, H. Schlarb (2022)
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A Neural Network Model of a Quasi-Periodic Elliptically Polarizing Undulator in Universal Mode
Ryan Sheppard, Cameron Baribeau, Tor Pedersen, Mark Boland, Drew Bertwistle (2022)
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Physics-informed neural network method for modelling beam-wall interactions
Kazuhiro Fujita (2022)
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Anomaly Detection in Particle Accelerators using Autoencoders
Jonathan P. Edelen, Nathan M. Cook (2021)
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Input Beam Matching and Beam Dynamics Design Optimization of the IsoDAR RFQ using Statistical and Machine Learning Techniques
Daniel Koser, Loyd Waites, Daniel Winklehner, Matthias Frey, Andreas Adelmann, Janet Conrad (2021)
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Neural Networks for ID Gap Orbit Distortion Compensation in PETRA III
Bianca Veglia, Ilya Agapov, Joachim Keil (2024)
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Time-Delayed Koopman Network-Based Model Predictive Control for the FRIB RFQ
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Uncertainty Aware ML-based surrogate models for particle accelerators: A Study at the Fermilab Booster Accelerator Complex
Malachi Schram, Kishansingh Rajput, Karthik Somayaji Peng Li, Jason St. John, Himanshu Sharma (2022)
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Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)
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Uncertainty Quantification for Virtual Diagnostic of Particle Accelerators
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Adaptive Latent Space Tuning for Non-Stationary Distributions
Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)
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Adaptive deep learning for time-varying systems with hidden parameters: Predicting changing input beam distributions of compact particle accelerators
Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)
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Using LSTM recurrent neural networks for monitoring the LHC superconducting magnets
Maciej Wielgosz, Andrzej Skocze■, Matej Mertik (2017)
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Coincident Learning for Beam-based RF Station Fault Identification Using Phase Information at the SLAC Linac Coherent Light Source
Jia Liang, William Colocho, Franz-Josef Decker, Ryan Humble, Ben Morris, Finn H. O'Shea, David A. Steele, Zhe Zhang, Eric Darve, Daniel Ratner (2025)
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Using Convolutional Neural Networks to Accelerate 3D Coherent Synchrotron Radiation Computations
Christopher Leon, Petr M. Anisimov, Nikolai Yampolsky, Alexander Scheinker (2025)
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Explainable physics-based constraints on reinforcement learning for accelerator controls
Colen, Jonathan, Schram, Malachi, Rajput, Kishansingh, Kasparian, Armen (2025)
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Accelerator system parameter estimation using variational autoencoded latent regression
Mahindra Rautela, Alan Williams, Alexander Scheinker (2024)
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Long Short-Term Memory Networks for Anomaly Detection in Magnet Power Supplies of Particle Accelerators
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Leveraging Prior Mean Models for Faster Bayesian Optimization of Particle Accelerators
Tobias Boltz, Jose L. Martinez, Connie Xu, Kathryn R. L. Baker, Zihan Zhu, Jenny Morgan, Ryan Roussel, Daniel Ratner, Brahim Mustapha, Auralee L. Edelen (2025)
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Optimizing Dynamic Aperture Studies with Active Learning
D. Di Croce, M. Giovannozzi, E. Krymova, T. Pieloni, S. Redaelli, M. Seidel, R. Tomás, F. F. Van der Veken (2024)
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Neural Network Prior Mean for Particle Accelerator Injector Tuning
Connie Xu, Ryan Roussel, Auralee Edelen (2022)

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Applications of object detection networks at high-power laser systems and experiments
Jinpu Lin, Florian Haberstroh, Stefan Karsch, Andreas Döpp (2022)
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A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung
S. Di Carlo, G. Bonvicini, N. A. Althubiti, R. Ayad, E. De La Cruz-Burelo, I. Domínguez, B. O. El Bashir, H. Farhat, J. Flanagan, R. Gillard, S. Izaguirre Gamez, K. Kanazawa, K. Kumara, D. Liventsev, P. L. M. Podesta-Lerma, D. Ricalde-Herrmann, D. Rodriguez Perez, G. Tejeda-Muñoz, M. Tobiyama I. Heredia de la Cruz (2022)
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Optimizing a Superconducting Radiofrequency Gun Using Deep Reinforcement Learning
David Meier, Luis Vera Ramirez, Jens Völker, Bernhard Sick, Jens Viefhaus, Gregor Hartmann (2022)
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Uncertainty aware anomaly detection to predict errant beam pulses in the SNS accelerator
Willem Blokland, Pradeep Ramuhalli, Charles Peters, Yigit Yucesan, Alexander Zhukov, Malachi Schram, Kishansingh Rajput, Torri Jeske (2021)
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Adaptive Machine Learning for Time-Varying Systems: Low Dimensional Latent Space Tuning
Alexander Scheinker (2021)
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Fast, efficient and flexible particle accelerator optimisation using densely connected and invertible neural networks
Renato Bellotti, Romana Boiger, Andreas Adelmann (2021)
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Invertible Surrogate Models: Joint surrogate modelling and reconstruction of Laser-Wakefield Acceleration by invertible neural networks
Friedrich Bethke, Richard Pausch, Patrick Stiller, Alexander Debus, Michael Bussmann, Nico Hoffmann (2021)
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Improving Surrogate Model Accuracy for the LCLS-II Injector Frontend Using Convolutional Neural Networks and Transfer Learning
Lipi Gupta, Auralee Edelen, Nicole Neveu, Aashwin Mishra, Christopher Mayes, Young-Kee Kim (2021)
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A Novel Approach for Classification and Forecasting of Time Series in Particle Accelerators
Sichen Li, Mélissa Zacharias, Jochem Snuverink, Jaime Coello de Portugal, Fernando Perez-Cruz, Davide Reggiani, Andreas Adelmann (2021)
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Real-time Artificial Intelligence for Accelerator Control: A Study at the Fermilab Booster
Jason St. John, Christian Herwig, Diana Kafkes, Jovan Mitrevski, William A. Pellico, Gabriel N. Perdue, Andres Quintero-Parra, Brian A. Schupbach, Kiyomi Seiya, Nhan Tran, Malachi Schram, Javier M. Duarte, Yunzhi Huang, Rachael Keller (2021)
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Surrogate Modeling of the CLIC Final-Focus System using Artificial Neural Networks
J. Ogren, C. Gohil, D. Schulte (2021)
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Physics-Based Deep Neural Networks for Beam Dynamics in Charged Particle Accelerators
Andrei Ivanov, Ilya Agapov (2020)
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Introduction to Machine Learning for Accelerator Physics
Daniel Ratner (2020)
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Machine learning for design optimization of storage ring nonlinear dynamics
Faya Wang, Minghao Song, Auralee Edelen, Xiaobiao Huang (2019)
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Studies in Applying Machine Learning to LLRF and Resonance Control in Superconducting RF Cavities
Jorge Alberto Diaz Cruz, Sandra Biedron, Manel Martinez-Ramon, Salvador Sosa, Reza Pirayesh (2019)
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The model of an anomaly detector for HiLumi LHC magnets based on Recurrent Neural Networks and adaptive quantization
Maciej Wielgosz, Matej Mertik, Andrzej Skocze■, Ernesto De Matteis (2017)
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Machine learning for analysis of plasma driven Ion source
N. Joshi, O. Meusel, H. Podlech (2018)
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First Steps Toward Incorporating Image Based Diagnostics Into Particle Accelerator Control Systems Using Convolutional Neural Networks
A. L. Edelen, S. G. Biedron, S. V. Milton, J. P. Edelen (2016)
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Neural Networks for Modeling and Control of Particle Accelerators
A. L. Edelen, S. G. Biedron, B. E. Chase, D. Edstrom, S. V. Milton, P. Stabile (2016)
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Harnessing the power of gradient-based simulations for multi-objective optimization in particle accelerators
Rajput, Kishansingh, Schram, Malachi, Edelen, Auralee, Colen, Jonathan, Kasparian, Armen, Roussel, Ryan, Carpenter, Adam, Zhang, He, Benesch, Jay (2024)
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BOOSTR: A Dataset for Accelerator Control Systems
Diana Kafkes, Jason St. John (2021)
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Model-free and Bayesian Ensembling Model-based Deep Reinforcement Learning for Particle Accelerator Control Demonstrated on the FERMI FEL
Simon Hirlaender, Niky Bruchon (2022)
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Autonomous Control of a Particle Accelerator using Deep Reinforcement Learning
Xiaoying Pang, Sunil Thulasidasan, Larry Rybarcyk (2020)
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AI-Assisted Transport of Radioactive Ion Beams
Sergio Lopez-Caceres, Daniel Santiago-Gonzalez (2025)
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Bayesian Optimization Algorithms for Accelerator Physics
Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lubsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez,

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SPIRAL2 Cryomodules Models: a Gateway to Process Control and Machine Learning
Adrien Vassal, Adnan Ghribi, François Millet, François Bonne, Patrick Bonnay, Pierre-Emmanuel Bernaudin (2021)
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Predicting Beam Transmission Using 2-Dimensional Phase Space Projections Of Hadron Accelerators
Anthony Tran, Yue Hao, Brahim Mustapha, Jose L. Martinex Marin (2022)
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Machine Learning for Orders of Magnitude Speedup in Multi-Objective Optimization of Particle Accelerator Systems
Auralee Edelen, Nicole Neveu, Yannick Huber, Mattias Frey, Christopher Mayes, Andreas Adelmann (2020)
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Machine learning assisted non-destructive transverse beam profile imaging
Zhanibek Omarov, Selcuk Haciomeroglu (2021)
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GAIA: A General AI Assistant for Intelligent Accelerator Operations
Frank Mayet (2024)
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AI-Enabled Operations at Fermi Complex: Multivariate Time Series Prediction for Outage Prediction and Diagnosis
Jain, Milan, Mutlu, Burcu O., Stam, Caleb, Strube, Jan, Schupbach, Brian A., John, Jason M.St., Pellico, William A. (2025)
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Superconducting radio-frequency cavity fault classification using machine learning at Jefferson Laboratory
Chris Tennant, Adam Carpenter, Tom Powers, Anna Shabalina Solopova, Lasitha Vidyaratne, Khan Iftekharuddin (2020)
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Generative Adversarial Networks (GAN) for compact beam source modelling in Monte Carlo simulations
David Sarrut, Nils Krah, Jean-Michel Létang (2019)
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Machine learning applied to single-shot x-ray diagnostics in an XFEL
A. Sanchez-Gonzalez, P. Micaelli, C. Olivier, T. R. Barillot, M. Ilchen, A. A. Lutman, A. Marinelli, T. Maxwell, A. Achner, M. Agåker, N. Berrah, C. Bostedt, J. Buck, P. H. Bucksbaum, S. Carron Montero, B. Cooper, J. P. Cryan, M. Dong, R. Feifel, L. J. Frasinski, H. Fukuzawa, A. Galler, G. Hartmann, N. Hartmann, W. Helml, A. S. Johnson, A. Knie, A. O. Lindahl, J. Liu, K. Motomura, M. Mucke, C. O'Grady, J-E. Rubensson, E. R. Simpson, R. J. Squibb, C. Såthe, K. Ueda, M. Vacher, D. J. Walke, V. Zhaunerchyk, R. N. Coffee, J. P. Marangos (2016)
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Siyu Wang, Shengran Dai, Jianhui Jiang, Shuang Wu, Yufei Peng, Junbin Zhang (2024)
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Inverse Surrogate Model of a Soft X-Ray Spectrometer using Domain Adaptation
Enrico Ahlers, Peter Feuer-Forson, Gregor Hartmann, Rolf Mitzner, Peter Baumgärtel, Jens Viefhaus (2025)
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Online tuning and light source control using a physics-informed Gaussian process Adi
A. Hanuka, J. Duris, J. Shtalenkova, D. Kennedy, A. Edelen, D. Ratner, X. Huang (2019)
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A. Hanuka, C. Emma, T. Maxwell, A. Fisher, B. Jacobson, M. J. Hogan, Z. Huang (2020)
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Application of ensemble machine learning algorithms and filtering techniques in slow orbit feedback systems of electron storage rings
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Goetzke, Gesa, Plumley, Rajan, Hartmann, Gregor, Maxwell, Tim, Decker, Franz-Josef, Lutman, Alberto, Dunne, Mike, Ratner, Daniel, Turner, Joshua J. (2025)
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Jeske, Torri, Kasparian, Armen, Lawrence, David, Britton, Thomas, Schram, Malachi, Moran, Patrick, Fanelli, Cristiano, Guo, Jiawei, Jarvis, Naomi, Maxwell, James, Keith, Chris (2025)
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Klekots, D., Bezshyyko, O., Golinka-Bezshyyko, L., Kubytskyi, V., Chaikovska, I. (2024)
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Piekarski, Michal (2024)
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Lin, Lucy (2024)
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Di Croce, Davide, Giovannozzi, Massimo, Montanari, Carlo Emilio, Pieloni, Tatiana, Redaelli, Stefano, Van der Veken, Frederik F. (2024)
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Wan, Jinyu, Plastun, Alexander, Ostroumov, Peter (2024)
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Knight, Ian, Mustapha, Brahim (2024)
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